



Chemistry H Review Practice

Writer: Ethan Shi

Editor: Patrick

Finals Range: (Chapter 5 + Chapter 13)

Note: (Review Guide/Review Practices/Practice Set) are UNOFFICIAL sources of reference and do NOT include or hint at questions in the actual exam!

A. Multiple Choice Questions

1. Hess's Law is based on which thermodynamic principle?
A) Energy cannot be created or destroyed, only transferred.
B) Entropy always increases in a spontaneous process.
C) The enthalpy change of a reaction is independent of the pathway.
D) The Gibbs free energy must be negative for a reaction to be spontaneous.
2. If a system absorbs 250 J of heat and does 100 J of work on the surroundings, what is the change in internal energy (ΔU)?
A) +150 J
B) -150 J
C) +350 J
D) -350 J
3. Which of the following will have the highest boiling point? (Assume complete dissociation where applicable)
A) 0.1 m NaCl
B) 0.1 m glucose
C) 0.1 m CaCl_2
D) 0.1 m methane
4. A 25.0 g sample of metal at 100°C is placed in 50.0 g of water at 20°C . The final temperature is 25°C . What is the specific heat of the metal? (Specific heat of water = $4.18 \text{ J/g}^\circ\text{C}$)
A) $0.385 \text{ J/g}^\circ\text{C}$
B) $0.557 \text{ J/g}^\circ\text{C}$
C) $1.25 \text{ J/g}^\circ\text{C}$
D) $2.09 \text{ J/g}^\circ\text{C}$
5. If a gas expands against a constant external pressure of 2 atm from 5 L to 10 L, how much work is done? ($1 \text{ L}\cdot\text{atm} = 101.3 \text{ J}$)
A) $-10 \text{ L}\cdot\text{atm}$
B) $+10 \text{ L}\cdot\text{atm}$
C) -1013 J
D) $+1013 \text{ J}$

6. What is the molarity of a solution containing 0.5 moles of NaCl in 250 mL of water?

- A) 0.5 M
- B) 1.0 M
- C) 2.0 M
- D) 0.25 M

7. The First Law of Thermodynamics states that:

- A) Energy is conserved in an isolated system.
- B) Entropy always increases in a spontaneous process.
- C) Heat flows from cold to hot objects.
- D) The enthalpy of a reaction is path-independent.

8. The standard enthalpy of formation (ΔH°_f) of an element in its most stable form is:

- A) Always positive.
- B) Always negative.
- C) Zero.
- D) Depends on temperature.

9. Which of the following is **least soluble** in water?

- A) CH₃OH (methanol)
- B) C₆H₆ (benzene)
- C) NaCl
- D) NH₃

10. Henry's Law states that the solubility of a gas in a liquid is directly proportional to:

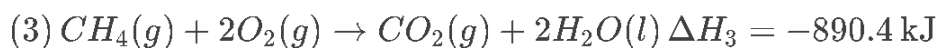
- A) The temperature of the liquid.
- B) The partial pressure of the gas above the liquid.
- C) The volume of the liquid.
- D) The density of the gas.

11. Why does NaCl dissolve well in water, but CCl₄ does not?

- A) NaCl is nonpolar, while CCl₄ is polar.
- B) NaCl dissociates into ions that interact with water's dipole.
- C) CCl₄ forms hydrogen bonds with water.
- D) Water has a lower dielectric constant than CCl₄.

12.

Question: Given:



Calculate the standard enthalpy of formation (ΔH_f) of $CH_4(g)$.

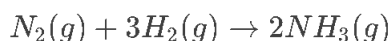
- A) -74.7 kJ/mol
- B) +74.7 kJ/mol
- C) -156.3 kJ/mol
- D) +156.3 kJ/mol

13. The Henry's Law constant (kH) for O₂ in water is $1.3 \times 10^{-3} \text{ M/atm}$. If the partial pressure of O₂ in air is 0.21 atm, what is the molar solubility of O₂ in water?

- A) $2.73 \times 10^{-4} \text{ M}$
- B) $6.19 \times 10^{-3} \text{ M}$
- C) $1.3 \times 10^{-3} \text{ M}$
- D) 0.21 M

14.

Question: Calculate ΔH for the reaction:



Given bond energies: $N \equiv N = 945 \text{ kJ/mol}$, $H - H = 436 \text{ kJ/mol}$, $N - H = 391 \text{ kJ/mol}$.

- A) -93 kJ
- B) +93 kJ
- C) -1476 kJ
- D) +1476 kJ

15. A 50.0 g block of iron ($c=0.449 \text{ J/g}^\circ\text{C}$) at 100°C is placed in 100 g of water ($c=4.18 \text{ J/g}^\circ\text{C}$) at 20°C . What is the final temperature?

- A) 24.1°C
- B) 30.2°C
- C) 60.0°C
- D) 75.8°C

B. True or False Questions

1. The standard enthalpy of formation (ΔH°_f) of ozone ($\text{O}_3(\text{g})$) is zero because it is a pure element.
2. If a solute dissolves endothermically ($\Delta H > 0$), its solubility will always decrease with increasing temperature.
3. The enthalpy changes for the reaction $2\text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{H}_2\text{O}(\text{g})$ is the same as the enthalpy of formation (ΔH°_f) of $\text{H}_2\text{O}(\text{g})$
4. Increasing temperature (in almost all cases) decreases gas solubility in liquids because k_H increases with temperature.
5. Using bond enthalpies to calculate ΔH of a reaction gives the exact value, even for reactions in solution.

C. Short Answer Questions (Calculations)

Given the enthalpy changes of the following reactions, calculate the enthalpy change for the reaction: $\text{A} + \text{B} \rightarrow \text{C} + \text{D}$.

- $\text{A} + \text{E} \rightarrow \text{C} + \text{F}; \Delta H_1 = -100 \text{ kJ/mol}$
- $\text{B} + \text{F} \rightarrow \text{D} + \text{E}; \Delta H_2 = 50 \text{ kJ/mol}$

Given the standard enthalpies of formation for the following substances, calculate the enthalpy change for the reaction: $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$.

- ΔH_f° of $\text{CH}_4(\text{g}) = -74.8 \text{ kJ/mol}$
- ΔH_f° of $\text{O}_2(\text{g}) = 0 \text{ kJ/mol}$
- ΔH_f° of $\text{CO}_2(\text{g}) = -393.5 \text{ kJ/mol}$
- ΔH_f° of $\text{H}_2\text{O}(\text{l}) = -285.8 \text{ kJ/mol}$